

MFU Access Points Energy Saving

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Abstract— In the contemporary technological landscape, access point antennas at Mae Fah Luang University experience excessive utilization owing to the manual control of their activation. Frequently, these access points remain powered despite minimal or no device connection requests, leading to unnecessary energy consumption and financial expenditure.

To curtail this inefficiency, the "MFU ACCESS POINTS ENERGY SAVING" system emerges as a compelling solution. The primary objective of this system is to autonomously manage access point activation based on available bandwidth. When the system detects an absence of devices or a minimal number of devices within an area, it orchestrates the deactivation of the 5G network. Leveraging the nearby 2.4G network as a detection mechanism, this approach optimizes access point usage, ensuring that resources are employed judiciously within Mae Fah Luang University.

This system seeks to address two crucial facets: comprehensively optimizing electricity consumption and promoting financial savings. By ensuring a more nuanced and responsive control mechanism for access points, it endeavors to minimize energy wastage during periods of minimal device connectivity, thus directly translating to substantial cost savings for the university.

Keywords— Access points, Comprehensive use of electricity, Money saving

I. INTRODUCTION

In the bustling environment of Mae Fah Luang University, the utilization of access point antennas has been extravagant, often resulting from the manual activation and deactivation process. Frequently, these access points cater to sparse device densities in specific times and areas, sometimes remaining entirely dormant with no connection requests. This misuse becomes a significant drain on both energy resources and finances, undermining the university's sustainability initiatives and operational efficiency.

To counteract this inefficiency, the MFU ACCESS POINTS ENERGY SAVING system has been conceived and developed as a strategic solution. The primary objective of this innovative system is to autonomously manage the activation and deactivation of access points, leveraging an intelligent algorithm that considers bandwidth availability. Operating on a sophisticated protocol, the system intelligently triggers the deactivation of the 5G network in instances where no devices are detected in a given area. Moreover, it switches to a power-saving mode when a minimal number of devices can seamlessly connect to nearby access points through the 2.4G network, effectively optimizing the utilization of access point antennas across the university's premises.

This initiative stands as a testament to Mae Fah Luang University's commitment to technological advancement and sustainable practices. By introducing an automated system that dynamically controls access point functionality based on real-time demand and network availability, the university aims to significantly reduce unnecessary energy consumption, lower operational costs, and foster an environment conducive to both academic and environmental excellence.

II. BACKGROUND AND RELATED WORK

A. Problem Statement

Extravagant use of access point antennas due to manual control, resulting in energy and monetary waste.

B. Proposed Solution

The system automates the activation/deactivation of access points based on sufficient bandwidth and device occupancy, specifically shutting down the 5G network if devices are absent or can connect to nearby access points via the 2.4G network.

C. Energy-Saving System Functionality

Automatic control of 5.0G network activation/deactivation based on access point occupancy using 2.4G network as a detector.

D. Networking Devices

Overview of essential hardware like laptops, routers, and access points crucial to network operation.

E. Software Tools

Description of key software tools: VMware Workstation for virtualization, Visual Studio for development, and Ubuntu as an open-source operating system.



Visual Studio



Ubuntu

III. ANALYSIS AND DESIGN

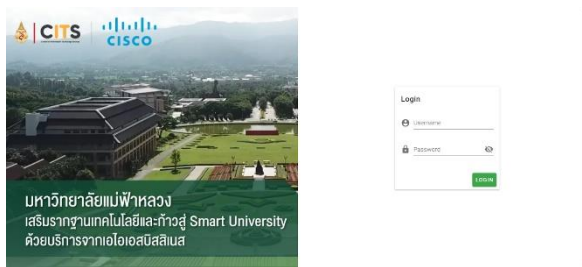
Block Diagram

- Illustrate the structure and functioning of the energy-saving system.

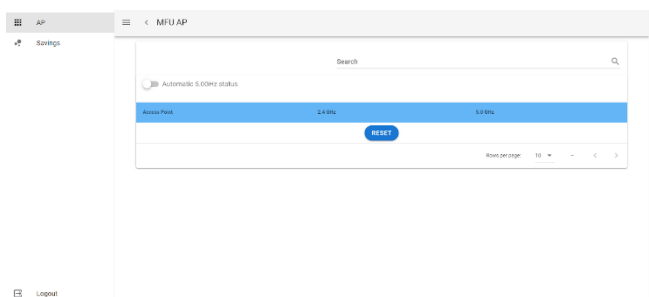
IV. EXPERIMENT AND RESULTS

- Website Design:

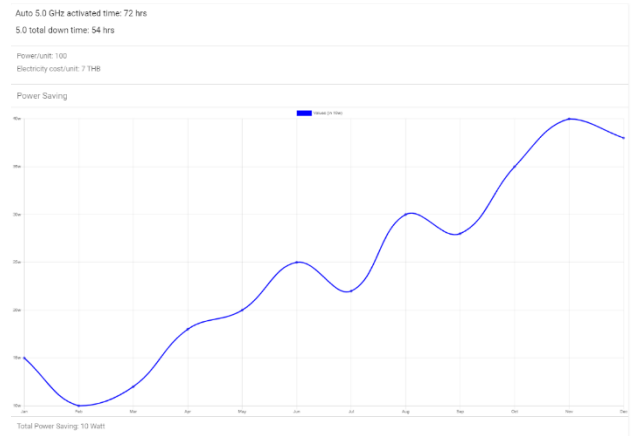
Discuss the design and purpose of login, AP, and savings pages.



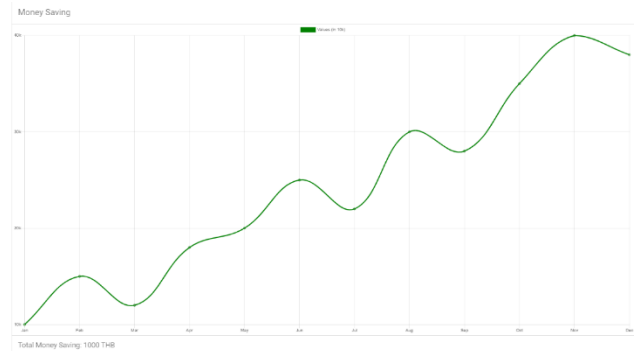
Login Page



Page



Savings Page



- Experiment:

Describe the testing process, including activating and observing the 'Automatic 5.0GHz Status' on the AP page.

- Experiment Results:

Present the outcomes of the system testing, including energy and cost savings data.

Day	Energy Savings	Money Savings
1		
2		
3		

Result of Experiment

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